

Secure Multiparty Computation Meets Deep Learning

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Preface

This is a Quarto book. To learn more about Quarto books visit <https://quarto.org/docs/books>.

Contents

The quarto book contains an organized structure of notes on a variety of secure multiparty computation protocols, implementation details, and a discussion of v2x applications and relevant deep learning models. The authors hope that any readers find these resources useful.

Resources

The [matcha editor](#) is used to construct some of the mathematical diagrams shown in this book. In order to export a matcha diagram, you need to enter the full-screen mode for the diagram, and click on the “export” drop-down which becomes available. This will allow you to save the math diagram as a png image. We will make liberal use of these throughout the book, especially for explaining complex security primitives such as beaver’s triples and homomorphic encryption.

Part I

Secure Multiparty Computation Primitives

This section of the text discusses a number of secure multiparty computation protocols in-depth and provides sample implementation to get you started.

1 Beaver's Triples Explained

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In order to mask fixed-point multiplication inputs, we need to use a secure primitive known as Beaver's triples. These were originally introduced by Beaver (1991) in his paper: [Efficient Multiparty Protocols Using Circuit Randomization](#)

1.1 Generating Boolean Beaver's Triples

The first step in masking the AND operation is to generate Boolean Beaver's Triples. These are random bits a , b , and c such that $c = a \cdot b$, where $\cdot$ represents the AND operation. Throughout this tutorial we define $a = a_A \oplus a_B$, $b = b_A \oplus b_B$, $c = c_A \oplus c_B$, and $r = r_A \oplus r_B$.

1.1.1 Step 1: Generating Inputs

Suppose we have two parties, **Alice** and **Bob**. We begin by first generating a random pair (a_i, b_i) for each party where i corresponds to the party in question. We generate these from the field F_2^2 which simply means that we select two random bits from $\{0, 1\}$.

$$\begin{array}{cc} \text{Alice} & \text{Bob} \\ \mathbb{F}_2^2 \xrightarrow{\$} (a_A, b_A) & \mathbb{F}_2^2 \xrightarrow{\$} (a_B, b_B) \end{array}$$

Next, Bob and Alice each generate random masking bits r_A and r_B respectively.

$$\begin{array}{cc} \text{Alice} & \text{Bob} \\ \mathbb{F}_2 \xrightarrow{\$} r_A & \mathbb{F}_2 \xrightarrow{\$} r_B \end{array}$$

1.1.2 Step 2: Obtaining Shares of c

After generating the masking bits, Alice and Bob invoke an instance of $\binom{2}{1}$ -OT₂ (see Chapter 2), where Alice acts as the sender with inputs $(r_A, r_A \oplus a_A)$, and Bob acts as the receiver with input b_B . Bob obtains $b_B a_A \oplus r_A$.

Now, Alice and Bob invoke another instance of $\binom{2}{1}$ -OT₂. This time, Bob serves as the sender with inputs $(r_B, r_B \oplus a_B)$, and Alice serves as the receiver with input b_A . Consequently, Alice obtains $b_A a_B \oplus r_B$.

$$\begin{array}{ccc}
 \text{Alice} & & \text{Bob} \\
 \text{Snd} = (r_A, r_A \oplus a_A) & \xrightarrow{\binom{2}{1}\text{-OT}_2} & \text{Rcv} = b_B a_A \oplus r_A \\
 \text{Rcv} = b_A a_B \oplus r_B & \xleftarrow{\binom{2}{1}\text{-OT}_2} & \text{Snd} = (r_B, r_B \oplus a_B)
 \end{array}$$

Note that objective is for Alice and Bob to each hold shares of c , denoted as c_A and c_B , respectively, such that their combination satisfies the equation $c = ab$.

Alice had received $b_A a_B \oplus r_B$ from Bob, and she already has $b_A a_A \oplus r_A$. She computes $b_A a_B \oplus r_B \oplus b_A a_A \oplus r_A$, which simplifies to $b_A a \oplus r$. Alice sets $c_A = b_A a \oplus r$.

Similarly, Bob computes $b_B a_A \oplus r_A \oplus b_B a_B \oplus r_B$, which simplifies to $b_B a \oplus r$. Bob sets $c_B = b_B a \oplus r$.

$$\begin{array}{ccc}
 \text{Alice} & & \text{Bob} \\
 c_A = b_A a \oplus r & & c_B = b_B a \oplus r
 \end{array}$$

It is easy to see that $c = c_A \oplus c_B = b_A a \oplus r \oplus b_B a \oplus r = ab$.

Thus, we have shown how Boolean Beaver's triples are randomly generated such that $c = c_A \oplus c_B$.

1.2 Application of Random Beaver's Triples

Now that randomized triples (a_A, b_A, c_A) and (a_B, b_B, c_B) have been generated for Alice and Bob respectively, we can use them cooperatively compute the output of some multiplication $x \cdot y$.

Suppose that Alice has shares of x and y (x_A, y_A) and Bob has shares (x_B, y_B) .

$$\begin{array}{ccc}
& \text{Alice} & \text{Bob} \\
f_A = y_A \oplus b_A & & f_B = y_B \oplus b_B \\
e_A = x_A \oplus a_A & & e_B = x_B \oplus a_B \\
& f = y \oplus b & \\
& e = x \oplus a &
\end{array}$$

Now that functional shares have been computed, we can use those to compute shares z_A and z_B for collectively obtaining the value of $x \cdot y$

$$\begin{array}{ccc}
& \text{Alice} & \text{Bob} \\
z_A = f \cdot a_A \oplus e \cdot b_A \oplus c_A & & z_B = f \cdot a_B \oplus e \cdot b_B \oplus c_B
\end{array}$$

Note: in the computation above, we approach the binary case of multiplication. Thus, the $\cdot$ operation is equivalent to the $\cap$ operation

We **XOR** these shares z_A and z_B together to get the following

$$\begin{aligned}
z_A \oplus z_B &= [(y \oplus b) \cdot (x \oplus a) \oplus (y \oplus b) \cdot a] \oplus [(x \oplus a) \cdot b \oplus c] \\
&= (y \oplus b)(x \oplus a) \oplus xb \oplus ab \oplus c \\
&= (y \oplus b)x \oplus xb \\
&= yx \oplus xb \oplus xb = yx
\end{aligned}$$

2 Oblivious Transfer

2.1 What is Oblivious Transfer?

If you're not familiar with Boolean operation XOR, please reference the following truth table:

A	B	$A \oplus B$
0	0	0
1	0	1
0	1	1
1	1	0

In standard computing practice, we often run into the issue of wanting to transfer values between parties. This problem can be modeled as follows:

Suppose that Alice has a set of values $\{s_1, s_2, \dots, s_n\}$ and Bob holds a selection bit i . Bob would like to obtain the value of s_i from Alice.

Without accounting for security, this can easily be accomplished by Bob sending his selection bit to Alice, and Alice sending the corresponding value s_i to Bob or Alice sending her entire set of values to Bob, and Bob picking the desired value s_i from the obtained set.

However, in Secure MPC (multi-party computation) we are concerned with the security of this process. More specifically, it is important that *Bob doesn't learn anything about Alice's s_i 's other than the one he receives* and *Alice doesn't learn anything about Bob's selection bit i* . Oblivious transfer is the secure mechanism by which we achieve this outcome. This is an extremely important Secure MPC tool because it has been proven that **OT is complete for all Secure MPC operations** (see O. Golrich (1987) and Killian (1988)).

As such, OT forms the basis for many secure computation protocols which have been developed in the past 40 years or so.

2.2 Extending Oblivious Transfer

Due to the the time complexity associated with Oblivious Transfer, researchers have been motivated to search for efficiency gains. Towards this end, several important developments have been made over time:

- It was demonstrated by R. Impagliazzo (1989) that if a reduction of OT to one-way-functions (black-box reduction) could be performed, it would imply that $P \neq NP$ (polynomial-class algorithm problems and NP class problems are not complexity-equivalent)
 - in other words, OT is harder than one-way-functions which form the basis of symmetric key encryption. See Chapter 3 for more information on this (OWF) one-way-functions.
- Beaver demonstrated one could extend a series of small base OT's to a large number of OT's using ONLY OWF's Beaver (1996)
- IKNP OT - (proposed in Y. Ishai (2003))
 - uses ROM (random oracle model) and symmetric key primitives to increase security and improve efficiency of OT
- Silent OT (Ferret) - (proposed in K. Yang (2020))
- Exponential Correlated OTs based on LPN Assumption (proposed in E. Boyle (2020))

2.3 Fundamentals of Oblivious Transfer

2.3.1 Random Oblivious Transfer (ROT)

what is the difference between ROT_l^m and OT_l^m ?

Random Oblivious Transfer is the most general form of OT. In this setting, we assume that there are two parties involved: Alice and Bob. Here, Alice functions as the *sender* and Bob functions as the *receiver*. Additionally, we assume a 1-out-of-2 ROT functionality. Alice and Bob follow the ROT_m^l protocol below:

The receiver (Bob) provides its selection bit b to the protocol, and two random strings are generated by the ROT protocol. These are r_0 and r_1 , one of which corresponds to the input bit b . The protocol provides these two random strings to the sender, and provides the receiver with a bit $r_b \in \{r_0, r_1\}$ which corresponds to the receiver's selection bit.*

How are bits selected by the protocol without assistance of a TTP?

Case 1: Bob uses the same selection bit b

Once the random outputs have been generated for Alice and Bob, they collectively perform an instance of $\binom{2}{1}$ -OT.

1. Alice inputs chosen bits (x_0, x_1) and Bob inputs selection bit b
2. Alice sends to Bob $(x_0 \oplus r_0, x_1 \oplus r_1)$
3. Bob performs $(x_b \oplus r_b) \oplus r_b$ to obtain x_b

Here, only one message is passed from Alice to Bob $S \rightarrow R$

Case 2: Bob chooses a different selection bit c

1. Bob sends to Alice $k = b \oplus c$ where b was the chosen bit input to the ROT protocol, and c is the chosen bit of the $\binom{2}{1}$ -OT
2. Alice permutes k with r_0 and r_1 to get r_k and $r_{\bar{k}}$.
3. Alice sends $(y_0, y_1) = (x_0 \oplus r_k, x_1 \oplus r_{\bar{k}})$ to Bob
4. Bob performs $y_c \oplus r_b$ to obtain x_c

2.3.2 Correlated Oblivious Transfer (COT)

Correlated Oblivious transfer has functionality defined as follows:

- The sender (Alice) has values x_0, x_1 such that the $\oplus$ of these values together results in δ
- Alice inputs Δ and receives r (a random l -bit string) and $r \oplus \Delta$.
- The receiver (Bob) obtains $r \oplus b\Delta$.

We use the notation COT_k^m to represent the correlated oblivious transfer functionality. Here m is the number of OT's to perform, and k is length of the random bit-strings used to mask the values exchanged.

In this setting, the ROT_l^m problem can be reduced to COT_k^m . If a random oracle is given, we can hash the output of a k length output to an l length output for ROT_l^m . However, it is not strictly necessary to use a random oracle. A correlation robust hash function may be used given that the joint distribution of hashes $(h(t_1 \oplus s), h(t_2 \oplus s), \dots, h(t_m \oplus s))$ for strings $s, t_1, \dots, t_m$ given $t_1, \dots, t_m$ is pseudo-random.

2.3.3 IKNP

3 One-Way-Functions

4 CrypTen

CrypTen is a secure multiparty computation framework introduced by B. Knott (2022). CrypTen is significant because it takes concepts which are well-established within the context of multiparty computation, and establishes an easy-to-use system which will feel familiar to researchers already know PyTorch. This is extremely important since the widespread adoption of secure computation techniques up to this point have largely been limited to (1) the complexity of primitives associated and (2) the low-level implementation used by most state-of-the-art systems such as Delphi and Cheetah.

4.1 Encrypting Models with CrypTen

In other systems such as Delphi, model encryption is treated as an issue of

Part II

Deep Learning

This portion of the text deals with deep learning and artificial intelligence applications. Specifically, it covers topics such as object detection and classification with computer vision, neural network primitives, and others. Other machine learning applications and import resources may also be included in this portion of the text

5 Recurrent Neural Networks

In order to work with time series data to predict the following point, we need to use a model which account for a varying amount of inputs. For example, in stock market price prediction, we want to inputs with a varying amount of input values available for each.

Just like other neural network architectures, RNNs take an input and feed it through an activation. However, RNNs loop the input multiple times through the activation in order to obtain an output.

Alternatively, we can think of this process as unrolling the network for sequential time series points.

Suppose:

$$\begin{aligned}t_1 &\rightarrow ReLU(W_{11}t_1 + b_{11}) = y_1 \\t_2 &\rightarrow ReLU(W_{12}t_2 + b_{12} + y_1) = y_2 \\&\vdots \\W_{22}y_2 + b_2 &= \text{predicted value}\end{aligned}$$

Notice that the output for t_1 is essentially used as a bias for the second input.

6 Word Embeddings and Word2Vec

These notes are based on Josh Starmer's video [Word Embedding and Word2Vec, Clearly Explained!!](#) on YouTube.com.

We need a method for converting words into numeric values. Otherwise, we won't be able to utilize them within a machine learning algorithm. We also want a method for representing words that understand the similarity between similar words - and mitigate the distance between those representations (they should be close together in a vector space if their semantic meanings are similar)

A very simple approach to understanding this problem is as follows:

Suppose you are given two phrases: "Troll 2 is great!" and "Gymkata is great!". In this case, there are four separate words ("Troll 2" is treated as a single word). Then, connect each of the inputs to at least one activation function. The number of activation functions for each input is equivalent to the number of components to associate with each word. *The weights on the connection to the each activation function represents the embedding values for a particular word.* Initially, we use random weights, but backpropagation can be used to predict the prior word.

Our prediction task here is the word which directly follows the input word. So, if the input is "Troll 2", then we want "is" to be the output word (the strongest predicted value). Then, connect the activation functions to outputs, and add weights to those connections. Then, the outputs are run through the softmax classification function. Crossentropy loss can then be used for backpropagation.

Additional context is used to improve the embeddings learned

1. *Continuous Bag of words* - the inputs are the words which occur before and after the target word, and the output is the word which occurs between them. For example, in "Troll 2 is great!", we want to embed "is" using the context "Troll 2" and "great!"
2. *Skip Gram* - the word in the middle is used to predict the surrounding word.

In practice, more than one activation is used in order to train embeddings for words, and the training dataset will be much larger. This results in much larger embeddings for most words.

Negative Sampling

Negative sampling is used in order to improve the speed at which embeddings can be trained. Suppose we have the input

$$\langle \text{"A"} = 1, \text{"aardvark"} = 0, \dots, \text{"zigzag"} = 0 \rangle$$

Then, we can ignore all of the weights which come from each of the other words and only include weights for the word of interest. However, there will still be weights which connect the activation layer to the output layer for each word

7 Residual connections

This section discusses residual connections based on the information provided by Wong (2021) in his medium article

7.1 Motivation for Residual connections

Deep Neural Networks such as YOLO (see Section 8.2.4.3) allow for greater accuracy and performance. However, deep networks like this make it more difficult for the model to converge during training.

Residual connections help to make training networks easier.

7.2 Formulation

Residual connections - allow data to reach other parts of a sequential network by skipping layers

This flow is depicted well by the following image:

Steps:

1. Apply identity mapping to x - perform element-wise addition $F(x) + x$: this is the **residual block**
 1. residual blocks may also include a ReLU activation applied to $F(x) + x$. *This works when dimensions of $F(x)$ and x are the same
 2. If dimensions of $F(x)$ and x are NOT the same, then you can multiply x by some matrix of constants W to scale it. $F(x) + Wx$

7.3 The Utility of Residual Blocks for Training Deep Networks

Empirical results have demonstrated that residual blocks increase the speed and ease of network convergence. There are a number of suspected reasons as to why this enables such performance gains.

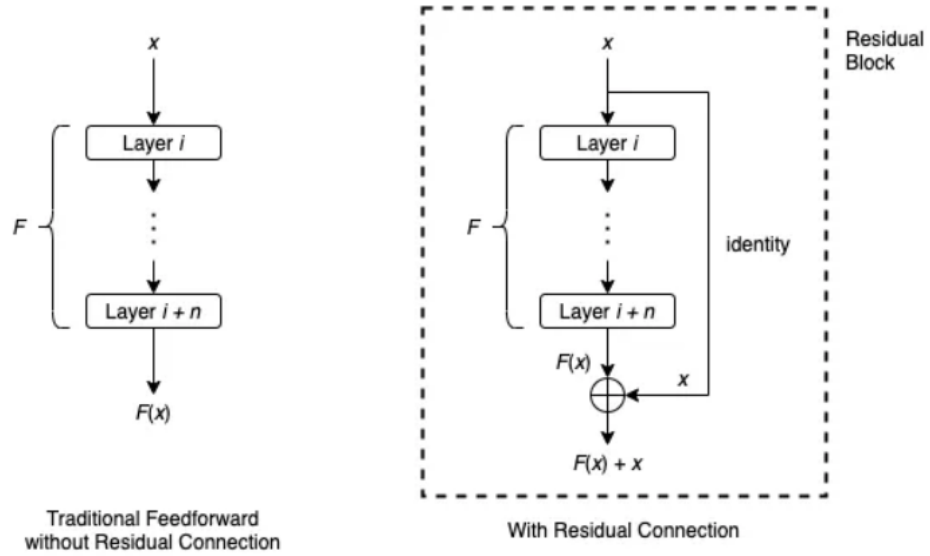


Figure 7.1: residual connection diagram

7.3.1 Ensemble of Shallow Neural Networks

7.3.2

8 Object Detection Algorithms

8.1 SSD: Single Shot MultiBox Detector

The single shot multibox detector is an algorithm presented by W. Liu (2016) for the purpose of taking a 300 x 300 input and generating bounding boxes on objects of interest within the image. The paper is linked [here](#)

8.2 A Comprehensive Review of YOLO (v1 to v8+)

J. Terven (2023) present review and analysis of the evolution of the Yolo algorithm, with a focus on the innovations and contributions made by each iteration, as well as the major changes in network architecture (and training tricks) which have been implemented over time.

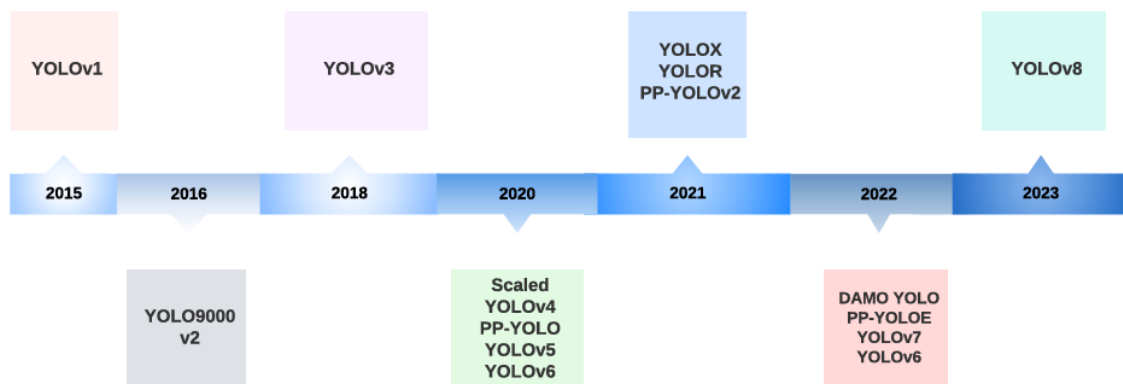


Figure 1: A timeline of YOLO versions.

Figure 8.1: A timeline of YOLO development

8.2.1 Applications of YOLO

Yolo has proven invaluable for a number of different applications

- autonomous vehicles
 - enables quick identification and tracking of objects like vehicles, pedestrians, bicycles and other obstacles
- action recognition
- video surveillance
- sports analysis
- human-computer interaction
- crop, disease, pest detection and classification
- face detection - biometrics, security, facial recognition
- cancer detection
- skin segmentation
- pill identification
- remote sensing
 - satellite and aerial imagery object detection / classification
 - land use mapping
 - urban planning
- security systems
- smart transportation systems
- robotics and drones

8.2.2 Evaluation Metrics

Average Precision (AP) and Mean Average Precision (mAP) are the most common metrics used in the object detection task. It measures average precision across all categories, providing a single value to compare different models

8.2.2.1 How AP works

- mAP is the average precision for accuracy of predictions across all classes of objects contained within an image
 - individual AP values are determined for each category separately.
- IOU (intersection over union)
 - measures the proportion of the predicted bounding box which overlaps which overlaps with the true bounding box

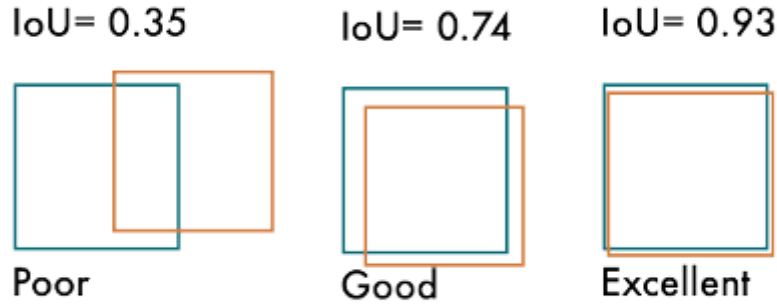


Figure 8.2: Intersection over union in practice

Different methods are used to compute AP when evaluating object detection methods on the COCO and VOC datasets (PASCAL-VOC)

8.2.3 Non-Maximum Suppression

A post-processing technique - reduces number of overlapping boxes and improves detection quality. Object detectors typically generate multiple bounding boxes around the same object. Non-max suppression picks the best ones and gets rid of the others.

The algorithm for this is defined below:

Algorithm 1 Non-Maximum Suppression Algorithm

Require: Set of predicted bounding boxes B , confidence scores S , IoU threshold τ , confidence threshold T

Ensure: Set of filtered bounding boxes F

```

1:  $F \leftarrow \emptyset$ 
2: Filter the boxes:  $B \leftarrow \{b \in B \mid S(b) \geq T\}$ 
3: Sort the boxes  $B$  by their confidence scores in descending order
4: while  $B \neq \emptyset$  do
5:   Select the box  $b$  with the highest confidence score
6:   Add  $b$  to the set of final boxes  $F$ :  $F \leftarrow F \cup \{b\}$ 
7:   Remove  $b$  from the set of boxes  $B$ :  $B \leftarrow B - \{b\}$ 
8:   for all remaining boxes  $r$  in  $B$  do
9:     Calculate the IoU between  $b$  and  $r$ :  $iou \leftarrow IoU(b, r)$ 
10:    if  $iou \geq \tau$  then
11:      Remove  $r$  from the set of boxes  $B$ :  $B \leftarrow B - \{r\}$ 
12:    end if
13:  end for
14: end while

```

Figure 8.3: Non-Max Suppression Alg

A useful visualization is also provided:



Figure 8.4: Non-Max Supression Vis

8.2.4 YOLO

The original authors of YOLO titled it as such for the reason that it only required a single pass on the image to accomplish the detection task. This is contrast to the other approaches used by Fast R-CNN and sliding window methods.

The output coordinates of the bounding box were detected using more straightforward regression techniques

8.2.4.1 YOLOv1

PASCAL-VOC AP: 63.4%

YOLOv1 predicted all bounding boxes simultaneously by the following process:

1. divide image into $S \times S$ grid
2. predict B bounding boxes of the same class and confidence for C different classes per grid element
3. each bounding box had five values:
 1. P_c - confidence score for the bounding box - how likely it contains an object and the accuracy of the box
 2. bx and by - coordinates of center of box relative to grid cell.
 3. bh and bw - height and width of box relative to full

4. output an $S \times S \times (B \times 5 + C)$ tensor
5. (optional) NMS used to remove redundant bounding boxes

Here is an example of that output:

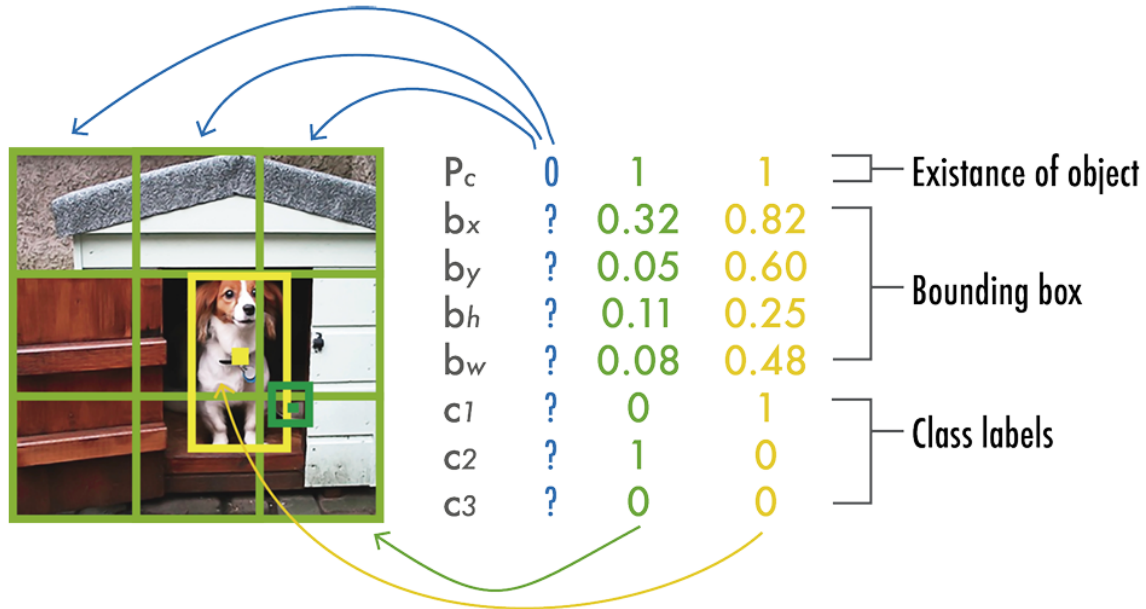


Figure 8.5: yolo output prediction

8.2.4.1.1 v1 Architecture

Normal Architecture

- 24 conv layers
 - 1×1 conv layers are used - reduce number of feature maps and keep parameters lower
 - leaky rectified linear unit activations
- 2 fc layers
 - predict bounding box coordinates / probs
 - linear activation function for final layer

FastYOLO

- Used 9 conv layers instead of 24 for greater speed (at the cost of reduced accuracy)

	Type	Filters	Size/Stride	Output
	Conv	64	$7 \times 7 / 2$	224×224
	Max Pool		$2 \times 2 / 2$	112×112
	Conv	192	$3 \times 3 / 1$	112×112
	Max Pool		$2 \times 2 / 2$	56×56
1×	Conv	128	$1 \times 1 / 1$	56×56
	Conv	256	$3 \times 3 / 1$	56×56
	Conv	256	$1 \times 1 / 1$	56×56
	Conv	512	$3 \times 3 / 1$	56×56
	Max Pool		$2 \times 2 / 2$	28×28
4×	Conv	256	$1 \times 1 / 1$	28×28
	Conv	512	$3 \times 3 / 1$	28×28
	Conv	512	$1 \times 1 / 1$	28×28
	Conv	1024	$3 \times 3 / 1$	28×28
	Max Pool		$2 \times 2 / 2$	14×14
2×	Conv	512	$1 \times 1 / 1$	14×14
	Conv	1024	$3 \times 3 / 1$	14×14
	Conv	1024	$3 \times 3 / 1$	14×14
	Conv	1024	$3 \times 3 / 2$	7×7
	Conv	1024	$3 \times 3 / 1$	7×7
	Conv	1024	$3 \times 3 / 1$	7×7
	FC		4096	4096
	Dropout 0.5			4096
	FC		$7 \times 7 \times 30$	$7 \times 7 \times 30$

Figure 8.6: yolo v1 architecture

8.2.4.1.2 v1 Training

Basic training process:

1. pretrain first 20 layers at resolution 224×224 with *ImageNet* dataset
2. add last four layers with randomly initialized weights - fine tune model with PASCAL VOC 2007 and PASCAL VOC 2012 at resolution 448×448

Loss functions:

- scaling factors
 - $\lambda_{coord} = 5$ - gives more weight to boxes with objects
 - $\lambda_{noobj} = 0.5$ - reduces importance of boxes with no object
- localization loss:
 - first two terms
 - computes error in predicted bounding box locations (x, y) and (w, h)
 - only penalizes boxes with objects in them
- confidence loss:
 - confidence error when object is detected (third term)
 - confidence error when no object is in box (fourth term)
- classification loss:
 - squared error of class conditional probabilities for each class if an object appears in the cell

8.2.4.2 YOLOv2 (YOLO 9000)

PASCAL-VOC AP: 78.6%

Improvements / Changes

1. Batch normalization - included on all convolutional layers
2. Higher resolution classifier - pretrained model (224×224) and then fine-tuned with ImageNet at a higher resolution (448×448) for ten epochs
3. fully convolutional - remove dense layers and use fully conv architecture
4. use anchor boxes to predict bounding boxes
 1. anchor box - box with predefined shapes for prototypical objects
 2. defined for each grid cell
 3. system predicts coordinates and class for every anchor box

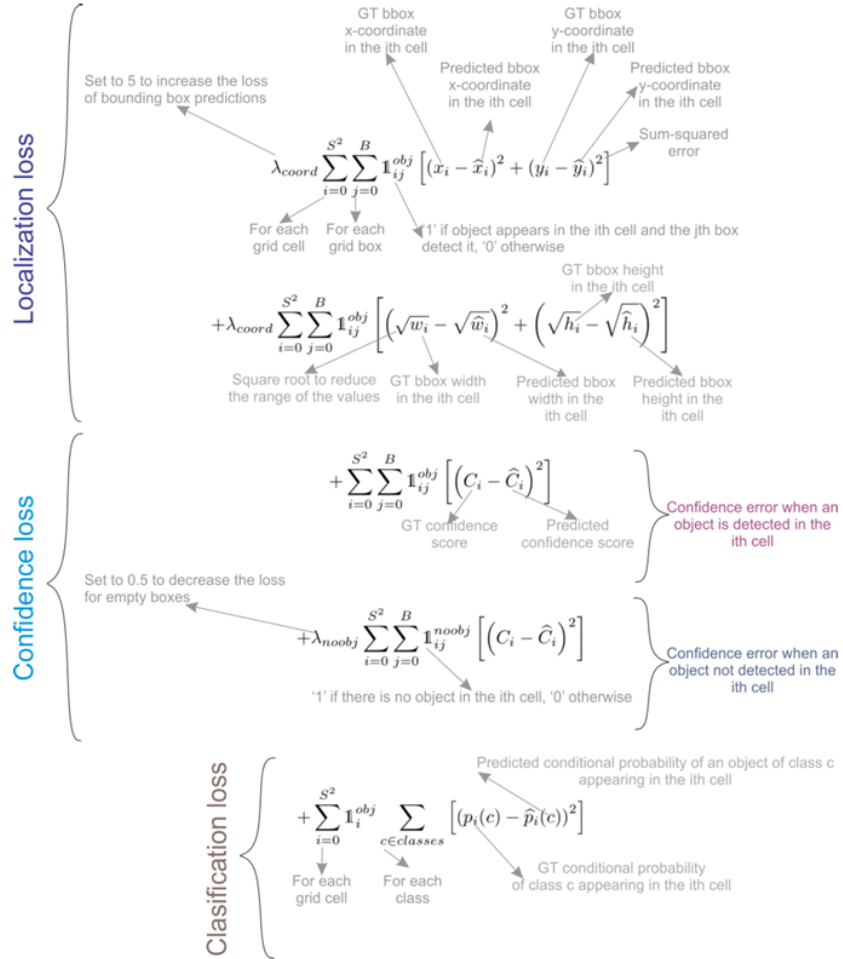


Figure 8.7: yolo v1 loss function

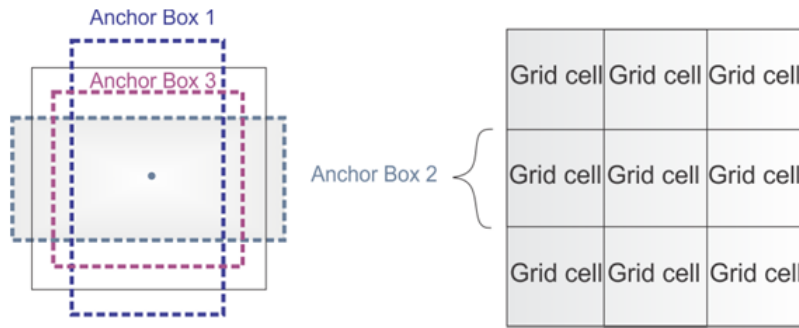


Figure 6: Anchor boxes. YOLOv2 defines multiple anchor boxes for each grid cell.

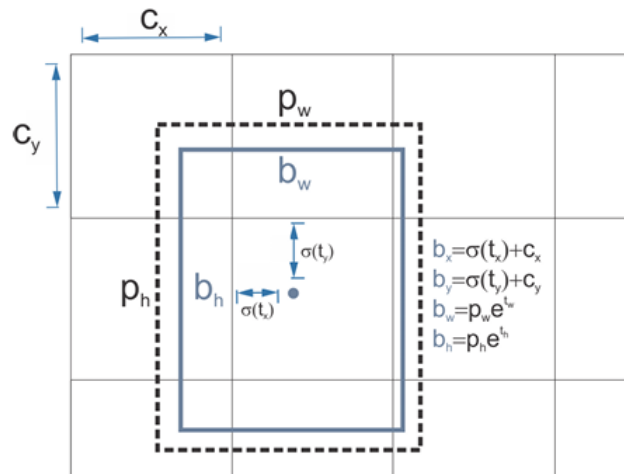


Figure 8.8: yolo v2 anchor boxes

5. Dimension clusters - pick good anchor boxes using k-means clustering on the training bounding boxes - improves accuracy of bounding boxes
6. Direct Location Prediction
7. Finer-grained features
 1. removed pooling layer - get feature 13 x 13 feature map for 416 x 416 images
 2. passthrough layer - 26 x 26 x 512 feature map -> stack adjacent features into different channels
8. Multi-scale training - train on different input sizes to make model robust to different input types

8.2.4.2.1 v2 Architecture

- backbone architecture -> Darknet-19
 - 19 conv layers
 - 5 max pool layers
 - * non-linear operation - uses OT to perform efficiently
 - use 1×1 conv between 3×3 to reduce parameters
 - batch normalization to help convergence
 - object classification head (replaces last 4 conv layers of YOLOv1)
 - * 1 conv layer (1000 filters)
 - * GAP layer
 - * Softmax classifier

8.2.4.3 YOLOv3

MS COCO AP: 36.2% AP(50): 60.6%

The code used to run YOLOv3 in Torch is provided at this [repository](#)

8.2.4.3.1 YOLOv3 Architecture

YOLOv3 makes use of a larger network architecture (backbone) called Darknet-53.

- replaces all max-pooling layers with **strided convolutions** and added residual connections (what are residual connections?) - see Chapter 7 for more information on this primitive.

The darknet architecture is presented here as well (visually):

Num	Type	Filters	Size/Stride	Output
1	Conv/BN	32	$3 \times 3 / 1$	$416 \times 416 \times 32$
2	Max Pool		$2 \times 2 / 2$	$208 \times 208 \times 32$
3	Conv/BN	64	$3 \times 3 / 1$	$208 \times 208 \times 64$
4	Max Pool		$2 \times 2 / 2$	$104 \times 104 \times 64$
5	Conv/BN	128	$3 \times 3 / 1$	$104 \times 104 \times 128$
6	Conv/BN	64	$1 \times 1 / 1$	$104 \times 104 \times 64$
7	Conv/BN	128	$3 \times 3 / 1$	$104 \times 104 \times 128$
8	Max Pool		$2 \times 2 / 2$	$52 \times 52 \times 128$
9	Conv/BN	256	$3 \times 3 / 1$	$52 \times 52 \times 256$
10	Conv/BN	128	$1 \times 1 / 1$	$52 \times 52 \times 128$
11	Conv/BN	256	$3 \times 3 / 1$	$52 \times 52 \times 256$
12	Max Pool		$2 \times 2 / 2$	$52 \times 52 \times 256$
13	Conv/BN	512	$3 \times 3 / 1$	$26 \times 26 \times 512$
14	Conv/BN	256	$1 \times 1 / 1$	$26 \times 26 \times 256$
15	Conv/BN	512	$3 \times 3 / 1$	$26 \times 26 \times 512$
16	Conv/BN	256	$1 \times 1 / 1$	$26 \times 26 \times 256$
17	Conv/BN	512	$3 \times 3 / 1$	$26 \times 26 \times 512$
18	Max Pool		$2 \times 2 / 2$	$13 \times 13 \times 512$
19	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
20	Conv/BN	512	$1 \times 1 / 1$	$13 \times 13 \times 512$
21	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
22	Conv/BN	512	$1 \times 1 / 1$	$13 \times 13 \times 512$
23	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
24	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
25	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
26	Reorg layer 17			$13 \times 13 \times 2048$
27	Concat 25 and 26			$13 \times 13 \times 3072$
28	Conv/BN	1024	$3 \times 3 / 1$	$13 \times 13 \times 1024$
29	Conv	125	$1 \times 1 / 1$	$13 \times 13 \times 125$

Figure 8.9: yolo v2 architecture

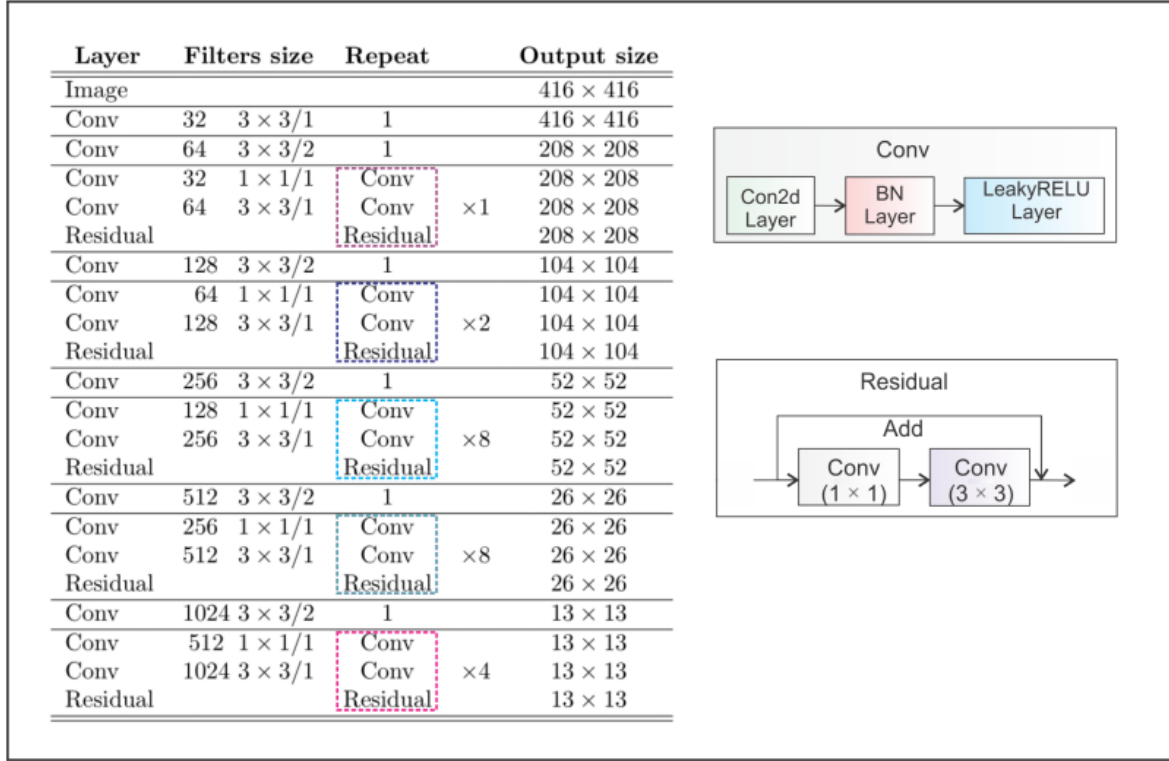


Figure 8.10: yolo v3 architecture

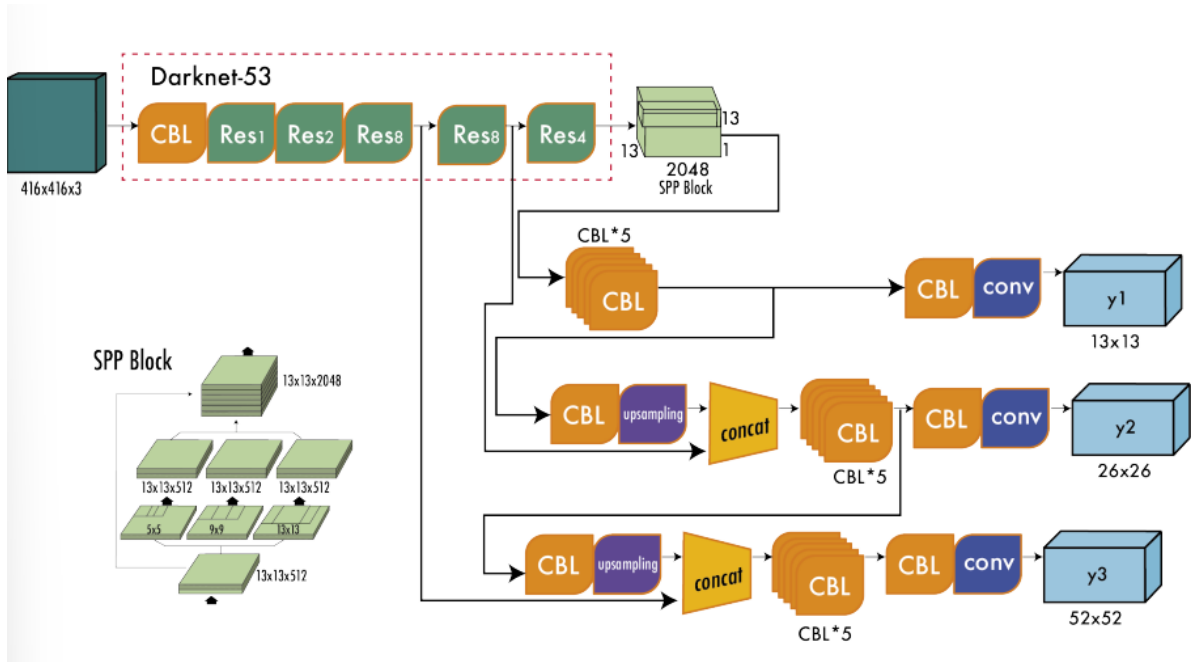


Figure 8.11: darknet 53 backbone

8.2.4.3.2 Multi-Scale predictions

- enables multi-scale predictions (predictions at multiple grid sizes)
- this helps to obtain finer detailed boxes (improves prediction of smaller boxes)
- YOLOv3 generates three separate outputs:
 - **y1**: 13×13 grid defines the output
 - **y2**: concatenating output after $(Res \times 4)$ with output of $(Res \times 8)$ - upsampling occurs from **y1** since the feature maps are of different sizes (13×13) and (26×26)
 - **y3**: upsample **y2** output to match 52×52 feature maps

8.2.4.3.3 Backbone, Neck, and Head

After release of YOLOv3, object detectors began to be described in terms of the backbone, neck, and head

Backbone

- Extracts useful features from the input image.
- A convolutional neural network trained on large-scale image classifications task (ImageNet)
- captures hierarchical features at different scales

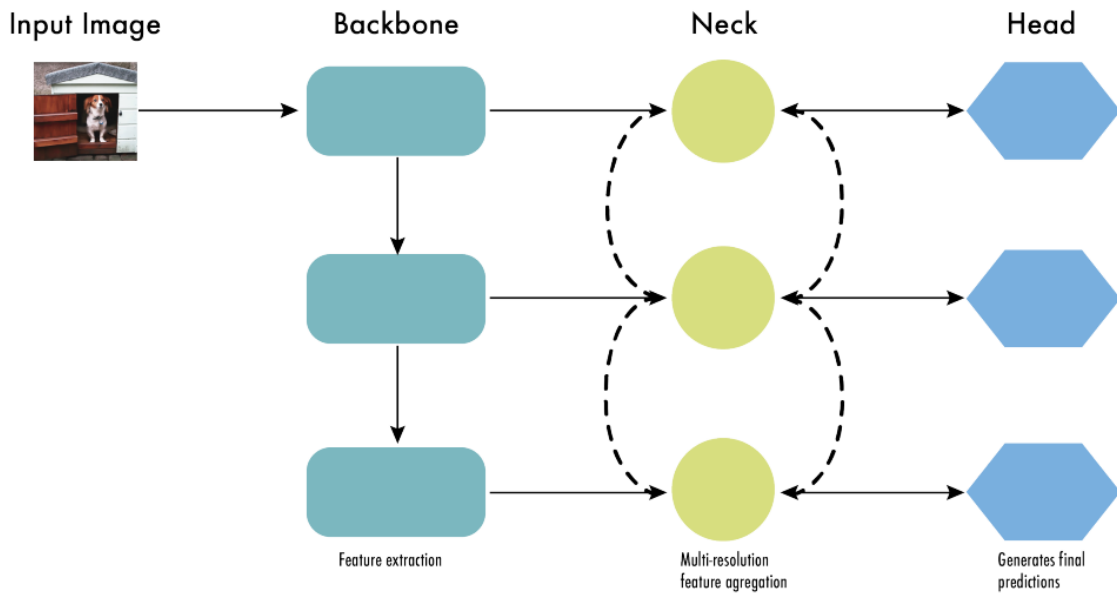


Figure 8.12: modern object detection architecture

- low-level features - earlier layers
- high-level features - deeper layers

Neck

- aggregates / refines features extracted by backbone
 - enhance spatial / semantic information across different scales
 - includes conv layers
 - includes feature pyramid networks

Head

- makes predictions based on features provided by backbone and neck
- consists of task-specific subnetworks to perform *classification*, *localization*, *localization*, *instance segmentation* and *pose estimation*
- non-maximum suppression filters out overlapping predictions (retains only most confident detections)

8.2.4.4 YOLOv4

MS COCO AP: 43.5% AP(50): 65.7%

The philosophy of YOLOv4 approaches optimization of the model into two categories: *bag of freebies* and *bag of specials*

Bag of Freebies:

- increase training time / cost
- do not affect inference time
- examples include *data augmentation*

Bag of Specials:

- increase inference time / cost
- improve accuracy of the model (MaP)
- examples include
 - enlarging receptive field ???
 - combining features ???
 - post-processing ???

8.2.4.4.1 Model Improvements

- **Enhanced network architecture with Bag-of-Specials**

- backbone: **Darknet-53** + Cross-stage Partial Connections (CSPNet) ?? + Mish Activation Function ??
 - * *tested several backbone architectures to choose best option*
 - * CSP reduces computation while maintaining accuracy
- neck: Spatial Pyramid Pooling (SPP)?? + **Multi-scale predictions** + modified path aggregation network PANet + modified spatial attention module (SAM)
 - * SPP increases receptive field without affecting inference speed
- detection head: **Same anchors as YOLOv3**

- **Advanced training with Bag-of-Freebies**

- regular augmentation
 - * random brightness, contrast, scaling, cropping, flipping, rotation
- special
 - * **mosaic augmentation:**
 - combines four images into a single one
 - reduces need for large mini-batches for batch normalization
 - * **DropBlock** regularization (instead of *Dropout*)
 - * **CIoU** loss and Cross mini-batch normalization **CmBN** for collecting statistics from the entire batch instead of just mini-batches

- these changes improve the **detector**
- **Self-adversarial Training**
 - improves model robustness to perturbations
- **Hyperparameter optimization with Genetic Algorithms**
 - genetic algs used on first 10% of periods
 - cosine annealing scheduler to alter learning rate during training

8.2.4.4.2 YOLOv4 Architecture

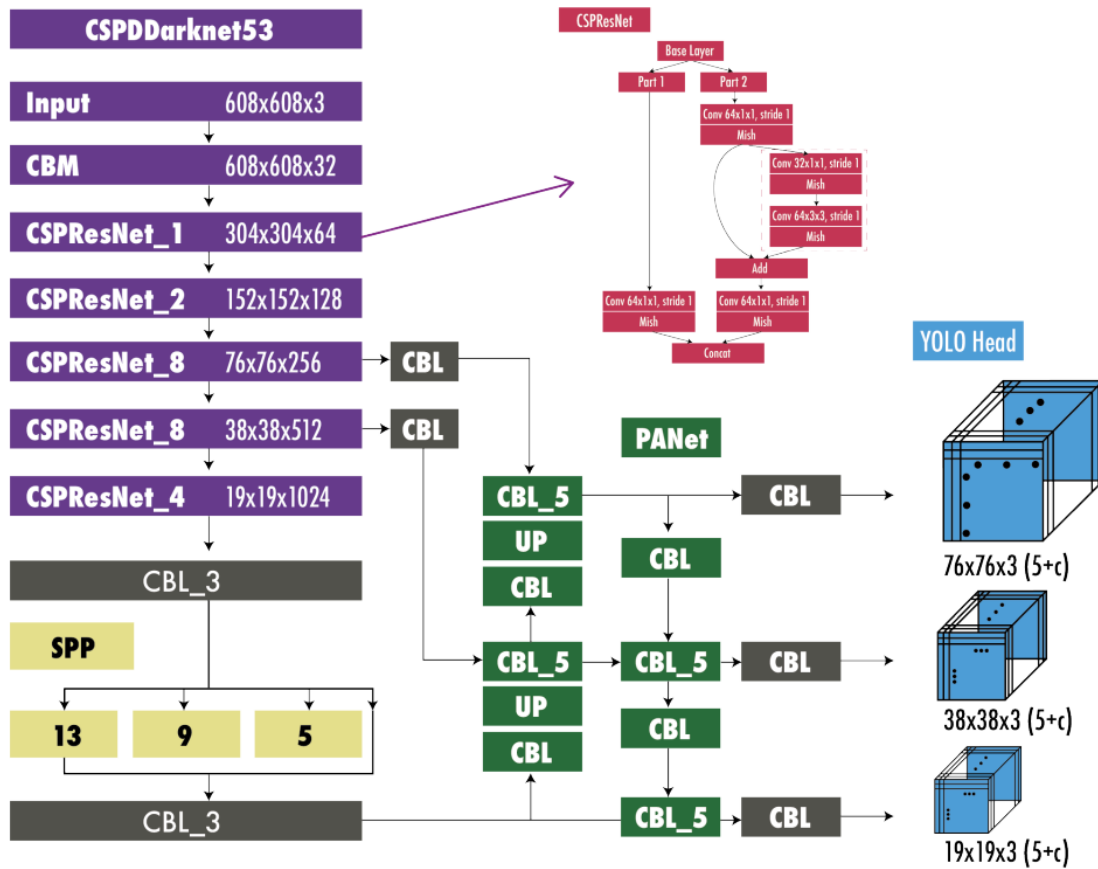


Figure 8.13: yolo v4 architecture

8.2.4.5 YOLOv5

YOLOv5x MS COCO AP: 50.7% AP(50): X%

No paper exists for YOLOv5 but there are several key advantages to using it

- It is developed in PyTorch instead of Darknet
- is open source and actively maintained by ultralytics
- is easy to use, train, and deploy
- many integrations for labeling, training, and deployment (mobile)

There are several scaled versions of this model:

- **YOLOv5n** (nano)
- **YOLOv5s** (small)
- **YOLOv5m** (medium)
- **YOLOv5l** (large)
- **YOLOv5x** (extra large)

This is useful for V2X privacy preserving applications because we will want to make use of smaller networks than might otherwise be used and compare their performance (efficiency) against larger networks.

This helps to answer the question of whether the accuracy tradeoff is worth the speedup.

8.2.4.5.1 Results

YOLOv5x achieves the following results

- AP 50.7% on COCO [batch size = 32, 640 pixels]
- AP 55.8% on COCO [1536 pixels]

8.2.4.6 Scaled-YOLOv4

YOLOv4-large MS COCO AP: 55.6%

Utilized scaling techniques (making larger to increase accuracy at the expense of speed, and scaling down increases speed at the expense of accuracy). Scaled-down models require less compute power and can run on embedded systems

Like YOLOv5, was developed in **PyTorch** instead of **Darknet**

8.2.4.6.1 Results

- YOLOv4-large - MS COCO AP: 56%
- YOLOv4-tiny - MS COCO AP: 22%

8.2.4.7 YOLOR

MS COCO AP: 55.4% AP(50): 73.3%

YOLOR stands for “you only learn one representation” and is novel for introducing multi-task learning approach which creates a single model for multiple tasks (classification, detection, pose estimation) by learning a general representation and using sub-networks to create task-specific representations!

8.2.4.7.1 Results

- YOLOR - MS COCO AP: 55.4% AP(50): 73.3%

8.2.4.8 YOLOX

MS COCO AP: 50.1%

Was designed off the back of the Ultralytics YOLOv3 (see Section 8.2.4.3) in PyTorch.

8.2.4.8.1 Model Changes and Improvements

- Anchor Free - simplified training and decoding process
- Multi Positives - center sampling (assigned center 3×3 area as positives) to account for imbalances produced by lack of anchors
- Decoupled Head - classification confidence and localization accuracy separated into two heads (connecting them leads to some misalignment)
 - **sped up model convergence and improved accuracy**
- Advanced Label Assignment - ambiguities were associated with ground truth bounding boxes (box overlap) - this is addressed with assigning procedure as Optimal Transport Problem . The authors develop a simplified version called simOTA
- Strong augmentations - MixUP and Mosaic augmentations were used

8.2.4.8.2 Architecture

8.2.4.9 YOLOv6

MS COCO AP: 52.5% AP(50): 70%

Adopted an anchor-free detector and provided a series of models at different scales for nuanced applications.

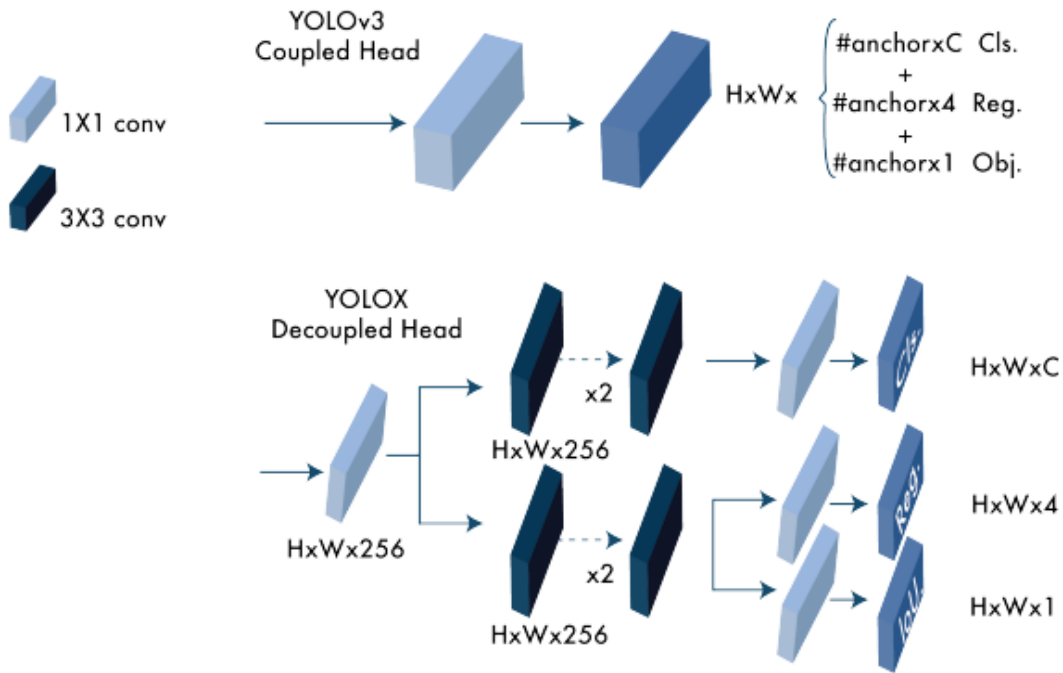


Figure 8.14: yolox architecture

8.2.4.9.1 Model Changes and Improvements

- New architecture backbone based on RepVGG . higher parallelism and use of neck based on RepBlocks or CSPStackRep Blocks and developed an efficient decoupled head
- Label Assignment with TOOD
- New classification and regression losses using VariFocal loss and SIOU / GioU regression loss
- self-distillation for regression and classification tasks
- quantization scheme for detection with RepOptimizer channel-wise distillation for a faster detector

8.2.4.10 YOLOv7

MS COCO AP: 55.9% AP(50): 73.5% - input 1280 pixels

Compared to YOLOv4 and YOLOR, YOLOv7 achieves a significant reduction in parameters used, and improves average precision by a meaningful increase in average precision as well.

8.2.4.10.1 Model Changes and Improvements

Architectural Changes

- Extended efficient layer aggregation network - ELAN allows for more efficient training and convergence
- Model Scaling for concatenation-based models - maintain optimal structure of the model by scaling the depth and width of the block with the same factor

Bag-of-Freebies

- Planned re-parameterized convolution - Architecture is inspired by RepConv . Identity connection outperforms residual in Resnet (Chapter 7) and concatenation in DenseNet
- Coarse label assignment for auxiliary head (training) and fine label assignment for the lead head (final out)
- Batch normalization in conv-bn-activation - integrates mean and variance of batch normalization into weight and bias of convolutional layer at inference stage (batch norm folding)
- Implicit knowledge (see Section 8.2.4.10)
- Exponential moving average as final inference model

8.2.4.11 DAMO-YOLO

MS COCO AP: 50.0%

Was inspired by a series of technologies relevant at the time and provided tiny/small/medium scaled model variants

- Neural architecture search (this was also used in Delphi) - MAE NAS -
- a large neck
- a small head
- aligned OTA label assignment - uses focal loss for classification cost and IoU of prediction / ground truth box as *soft label*. Enables selection of *aligned samples* for targets
- knowledge distillation
 - (1) teacher guiding student (stage 1)
 - (2) student fine-tuning (stage 2)
 - enhancements in distillation approach
 - * align module - adapts student features to same resolution as teacher's
 - * channel-wise dynamic temperature - normalizes teacher and student features

8.2.4.12 YOLOv8

YOLOv8x MS COCO AP: 53.9% (640 pixels)

A version of YOLO released by ultralytics which is anchor free and uses mosaic augmentation for training up to the last ten epochs. It provides five scaled versions

1. YOLOv8n (nano)
2. YOLOv8s (small)
3. YOLOv8m (medium)
4. YOLOv8l (large)
5. YOLOv8x (extra large)

8.2.5 PP-YOLO Models

The PP-YOLO models were developed in parallel with the standard YOLO variants. These models were based on YOLOv3 but were developed in the PaddlePaddle deep learning platform instead. The goal of their work was to show how an object detector should be constructed step-by-step, not to provide any novel functionality or a new approach.

8.2.5.1 PP-YOLO

MS COCO AP:45.9% AP(50): 65.2%

8.2.5.1.1 Divergence from YOLOv3

- ResNet50-vd backbone + deformable convolutions replace DarkNet-53 (Section [8.2.4.3](#))
 - achieves higher classification accuracy on Imagenet
- Larger batch size - improves training stability
- Maintained moving averages for trained parameters
- DropBlock applied on FPN
- IoU loss added with L1-Loss for bounding box regression
- IoU prediction branch for measuring localization accuracy (and optimization)
- Grid Sensitive Approach like YOLOv4 (Section [8.2.4.4](#)) - improves bounding box center prediction at grid boundary
- Matrix NMS (parallelized NMS for faster computation)
- CoordConv - 1×1 convolution of the FPN. Allows for learning translational invariance
- Spatial Pyramid Pooling increases receptive field of the backbone

8.2.5.1.2 PP-YOLO Augmentations and Preprocessing

- Mixup Training - weights $\sim \text{Beta}(\alpha = 1.5, \beta = 1.5)$
- Random color distortion
- Random expand
- Random crop and random flip with probability 0.5
- RGB channel z-score normalization $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$

8.2.5.2 PP-YOLOv2

MS COCO AP: 49.5%

8.2.5.2.1 Changes and Improvements

- Backbone changed from ResNet50 to ResNet101
- Path aggregation network instead of FPN
- Mish Activation function applied to the detection neck
- Larger input sizes - increases performance on small objects
- Modified IoU aware branch - soft label format instead of soft weight format

8.2.5.3 PP-YOLOE

MS COCO AP: 51.4% (78.1 FPS NVIDIA V100)

8.2.5.3.1 Changes and Improvements

- Anchor Free (following trends of other yolo models)
- New backbone and neck - modified neck with RepResBlocks (combine dense and residual connections)
- task alignment learning (see Section 8.2.4.8)
- efficient task-aligned head (ET-head) - single head based on TOOD instead of splitting the classification / detection heads like with YOLOX (Section 8.2.4.8)
- varifocal (VFL) and distribution focal loss (DFL)
 - VFL weights positive samples using a target score which places a greater importance on high-quality samples during training
 - DFL extends Focal Loss from discrete to continuous labels - better for representations which combine quality estimation and class prediction - this allows for better depiction of flexible distributions in real data (eliminates risk of inconsistency)

The authors provide several scaled models

- PP-YOLOE-s (small)
- PP-YOLOE-m (medium)
- PP-YOLOE-1 (large)
- PP-YOLOE-x (extra large)

8.2.6 Summary of YOLO

Table 4: Summary of YOLO architectures. The metric reported for YOLO and YOLOv2 were on VOC2007, while the rest are reported on COCO2017

Version	Date	Anchor	Framework	Backbone	AP (%)
YOLO	2015	No	Darknet	Darknet24	63.4
YOLOv2	2016	Yes	Darknet	Darknet24	63.4
YOLOv3	2018	Yes	Darknet	Darknet53	36.2
YOLOv4	2020	Yes	Darknet	CSPDarknet53	43.5
YOLOv5	2020	Yes	Pytorch	Modified CSP v7	55.8
PP-YOLO	2020	Yes	PaddlePaddle	ResNet50-vd	45.9
Scaled-YOLOv4	2021	Yes	Pytorch	CSPDarknet	56.0
PP-YOLOv2	2021	Yes	PaddlePaddle	ResNet101-vd	50.3
YOLOR	2021	Yes	Pytorch	CSPDarknet	55.4
YOLOX	2021	No	Pytorch	Modified CSP v5	51.2
PP-YOLOE	2022	No	PaddlePaddle	CSPRepResNet	54.7
YOLOv6	2022	No	Pytorch	EfficientRep	52.5
YOLOv7	2022	No	Pytorch	RepConvN	56.8
DAMO-YOLO	2022	No	Pytorch	MAE-NAS	50.0
YOLOv8	2023	No	Pytorch	YOLO v8	53.9

Figure 8.15: summary of yolo architectures and results

9 Implementation Guides for Yolo Object Detectors

Yolo5 is an extremely popular choice of object detector model. It has empirically demonstrated a high mean average precision, is efficient, supports an array of model scales, and is very easy to implement. Moreover, it is a torch port of the yolo family, which makes it even more popular among machine learning researchers and the like.

This guide provides a simple explanation for implementing the YOLOv5 model on a very simple example. Please note that this tutorial borrows heavily from the article written by Maindola (2021) . Practically speaking, the purpose of this entry is for ease of use in connecting to other useful references throughout the remainder of this book.

9.1 YOLOv5 Architecture In Depth

A comprehensive description of the YOLOv5 network architecture is provided at the [documentation source from ultralytics](#).

[Open Genus](#) also provides some good reference material on the architecture of the Yolo5 architecture.

This [youtube video](#) also provides a useful explanation for the partial connections structure of the YOLO model architecture.

Tensors are either added or concatenated together when they are passed through the network. YOLO makes exclusive use of tensor concatenation.

10 AdderNet

What is AdderNet? AdderNet is a novel approach to deep learning, specifically with convolutional neural networks which seeks to reduce computational complexity by replacing multiplication operations with additions. Namely, this is accomplished by supplanting convolution operations (cross-correlation) with l_1 -norm distance. This substitution can be made specifically because the cross-correlation is used to measure the similarity between the input feature and convolution filters. Thus, absolute value can also be used as such a measure.

Optimizations for the training of such adder networks have also been developed. Such improvements include enhanced back-propagation schemes with regularized gradients. This helps to ensure sufficient updates of templates and better network convergence

10.1 Defining Adder-Networks

Suppose the following setting:

- You have some filter $F \in \mathbb{R}^{d \times d \times c_{in} \times c_{out}}$
 - d = kernel size
 - c_{in} = input channel
 - c_{out} = output channel
- Input feature: $X \in \mathbb{R}^{H \times W \times c_{in}}$
- Output $Y(m, n, t) = \sum_{i=0}^d \sum_{j=0}^d \sum_{k=0}^{c_{in}} S(X(m+i, n+j, k), F(i, j, k, t))$

Here, S is the similarity measure. Adder-Networks use the l_1 distance as the measure of similarity between two points. Calculate the sum of the absolute differences of two points' vector representations (this does not contain any multiplications)

Formulation

The following presents a general form for a convolution-equivalent layer. Here $S(\cdot, \cdot)$ is a pre-defined similarity measure. When $S(x, y) = x \times y$, this is simply the convolution operation

$$Y(m, n, t) = \sum_{i=0}^d \sum_{j=0}^d \sum_{k=0}^{c_{in}} S(X(m+i, n+j, k), F(i, j, k, t)) \quad (10.1)$$

The absolute value (l1-norm) replaces the convolution operation (cross-correlation) as the measure of similarity between input features and the filter

$$Y(m, n, t) = - \sum_{i=0}^d \sum_{j=0}^d \sum_{k=0}^{c_{in}} |X(m+i, n+j, k) - F(i, j, k, t)| \quad (10.2)$$

10.1.1 Convolution v. L1-Distance

Differences are as follows:

- the convolution filter can exist as a weighted sum of values in the feature maps (can take a positive or negative value)
- adder filters are ALWAYS NEGATIVE
 - use batch normalization for help (normalize adder outputs to correct range)

Part III

V2X Applications

This portion of the text deals specifically with discussions of V2X applications and papers which have tackled these sorts of the problems in the past.

11 Red Light Violation Detection

11.1 Motivation

Secure Red Light Violation detection is an important application of secure machine learning protocols. Oftentimes, these systems will require the use of image segmentation and object recognition protocols. The images used for this task expose often-times sensitive data about individual users.

In the practical setting of interest, this means exposing license-plate information, associated vehicles, and location information available from the images themselves.

11.2 V2I Algorithms for RLR Detection

The authors of this paper have constructed V2I mechanisms for red light running (RLR) detection, wrong way entry (WWE), and an array of other import tasks in the context of V2X. See the citation Dokur and Katkoori (2022)

11.2.1 Red Light Violation Detection Algorithm

The proposed system utilizes the following logic to detect whether a car will violate a red light. A car which is approaching an intersection is connected to road-side units (RSUs) which are installed at traffic lights in an intersection.

Each light is said to be located at points $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$, $D(x_4, y_4, z_4)$ and $E(x_5, y_5, z_5)$ respectively.

Unlike image-based systems, this system assumes V2I communication between the traffic lights and the vehicle in question. This means that the traffic state does not need to be determined by an image classifier. Rather, we already have this information by default.

11.3 Thao et.al on Traffic Violation Detection

The paper proposed by L. Thao (2022) introduces a mechanism for detecting red light violations automatically. There paper is titled: *Automatic Traffic Red-Light Violation Detection Using AI*

11.3.1 Problem Setting

The reason AI technologies (image classification and detection) systems are better suited than standard sensor technologies is that they are able to operate more consistently, even when the number of vehicles in the setting increases dramatically.

11.3.2 System Design and Solution Approach

Separate the task into three parts:

1. vehicle violation detection
2. red signal change monitoring
3. vehicle recognition

Vehicle Violation Detection

The YOLOv5s pretrained model (COCO dataset) is used for detecting violating vehicles. After detecting violation, following frames are used to try and determine the license plate (identify vehicle). See Section 8.2.4.1 for more information on the YOLO object detection model.

Below, a picture of the overall system flow is presented:

- **vehicle tracking** - performed every 5 frames
 - if IOU (intersection over union) of bounding box is close to one from a previous frame, then the car is assumed to be the same one from that frame.
- **violation line detection**
 - image processing is used to determine traffic lines
 - boundary lines are drawn onto frames captured by the camera later
- **traffic state detection**
 - color filters and image processing used to detect changes in the state of the traffic light

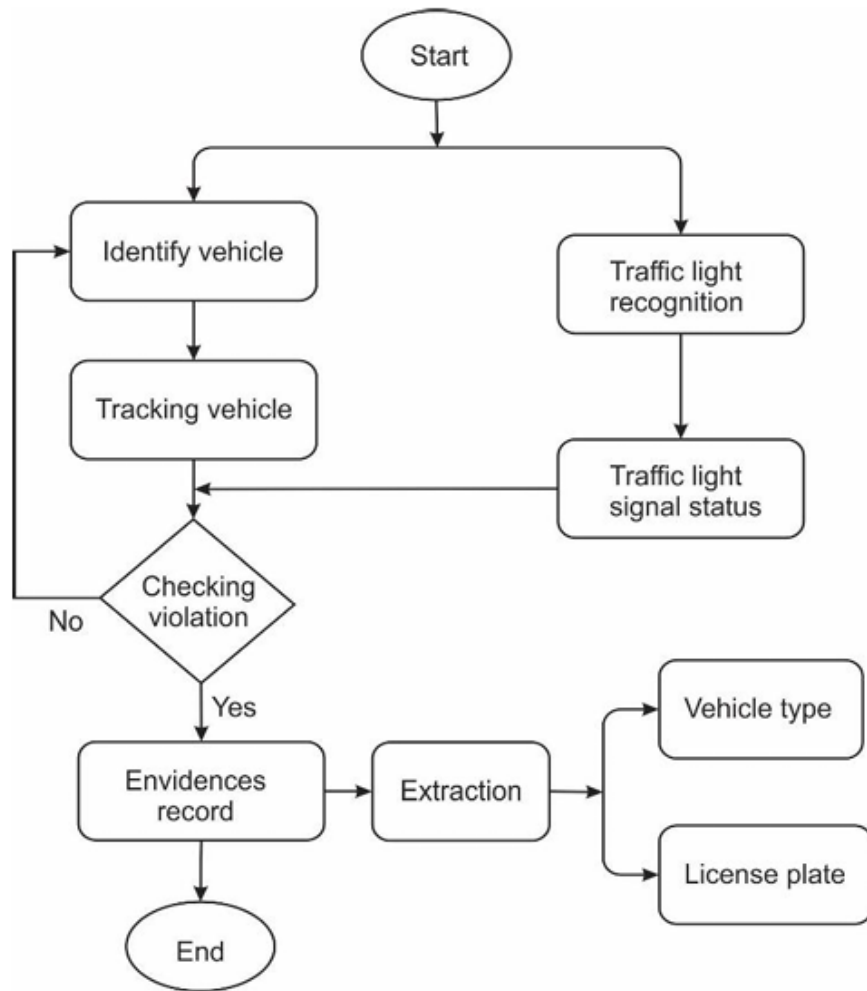


Figure 2. Algorithm of violation detection

Figure 11.1: system flow

11.3.3 Primary Contributions

1. Implementation of modified *YOLOv5s* model
 1. used parameter changes from original model
 2. achieved following accuracy results:
 1. 82% - vehicle identification
 2. 90% - traffic signal status change
 3. 86% - violation detection
2. Best Performing Architecture given below (v3 / v4)

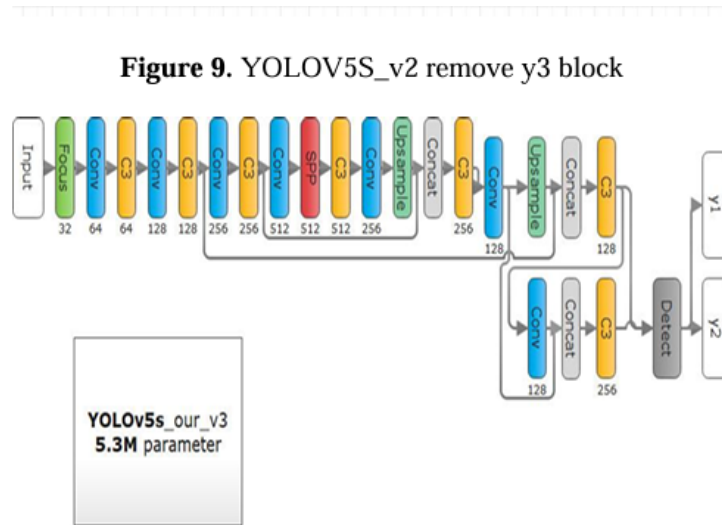


Figure 10. YOLOV5S_v3 reduce ½ filter in Conv, remove y3 block

Figure 11.2: modified Yolo architecture

According to a labmate, it may only be necessary to fine tune the YOLO model on 10 epochs. It may not be necessary to do any more than that.

11.4 Goma et.al on RLR Detection with SSD (Single Shot Detector)

Work by J. Goma (2020) demonstrates the ability to detect red-light-running and over-speeding with a high level of accuracy. Specifically, they achieve **100%** accuracy on red light running detection and **92.1%** accuracy for over-speeding viola-

tions. They accomplish these results using a CNN applied to an SSD (single deep neural network)

11.4.1 Methods and System Design

12 Traffic Flow Forecasting

12.1 Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting

In this paper, S. Guo (2019) proposed a method for traffic flow prediction which utilized an **attention based spatial-temporal graph convolutional network**. They aimed to model several time-dependencies: (1) recent, (2) daily-periodic, and (3) weekly-periodic dependencies. The *attention mechanism* captures spatial-temporal patterns in the traffic data and the *spatial-temporal convolution* is used to capture spatial patterns while *standard convolutions* describe temporal features.

12.1.1 Core Contributions of ASTGCN

Difficulties of the traffic forecasting problem

- it is difficult to handle unstable and nonlinear data
- prediction performance of models require extensive feature engineering
 - domain expertise is necessary
- cnn - spatial feature extraction from grid-based data, gcn - describe spatial correlation of grid based data
 - fails to simultaneously model spatial temporal features and dynamic correlations of traffic data
 - The convolution component of the network is meant to capture the neighborhood feature context surrounding each node - in a spatial temporal model, this means that a three dimensional kernel is required to account for the time dimension
 - the spatial-temporal influence of different edges / connections between nodes varies across different time points.

Addressing these issues:

- develop a *spatial-temporal attention mechanism*
 - learns dynamic spatial-temporal correlations of traffic data

- temporal attention is applied to capture dynamic temporal correlations for different times
- the attention mechanism learns the
- Design of *spatial-temporal convolution module*
 - has graph convolution for modeling graph structure
 - has convolution in temporal dimension (kind of like 3-d convolution)
 - the output of the model is a label (for the state of traffic flows) at each node in the traffic network

Related Work:

- two main types of convolutions for graph data
 - spatial methods - directly perform convolution filters on a graph's nodes and their neighbors
 - * Niepert et al. propose a heuristic linear method for selecting the neighborhood of every center node
 - spectral methods -

13 Summary

In summary, this book has no content whatsoever.

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